Project Report: Microcontroller-Based Voltage Protector Device

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Introduction

Modern electrical appliances are highly susceptible to damage from power grid anomalies, including sudden voltage surges, sags, and over-current faults. This project details the design, simulation, and validation of a Microcontroller-Based Voltage Protector Device capable of continuously monitoring grid parameters and automatically isolating appliances during fault conditions. The system features a custom hardware architecture paired with a Python-based Graphical User Interface (GUI) for real-time monitoring and threshold configuration.

System Architecture and Hardware Design

The core of the protection system is built around the **ATmega32 microcontroller**, which serves as the central processing unit handling data acquisition, logic processing, and peripheral communication.

To accurately simulate and measure the power grid parameters, the hardware design incorporates:

- **Power Grid Simulation & Voltage Sensing:** A simulated AC power grid is stepped down using a Voltage Transformer (VT). The stepped-down voltage is rectified and conditioned to safely interface with the microcontroller for real-time grid voltage measurement.
- **Current Sensing:** An **ACS712 Hall-effect current sensor** is integrated into the load path. This sensor provides an isolated, proportional analog voltage output corresponding to the current drawn by the protected appliance, ensuring precise load monitoring.

Software Implementation and Interrupts

The firmware is heavily interrupt-driven to ensure non-blocking, real-time responsiveness to both grid faults and user inputs. The system utilizes three primary hardware interrupts:

1. **ADC Interrupt:** Configured to handle all analog inputs. It continuously samples the conditioned signals from the Voltage Transformer and the ACS712 sensor, allowing the ATmega32 to calculate RMS voltage and current values without halting the main execution loop.

2. **Timer0 Interrupt:** Dedicated to scanning and debouncing the physical push buttons on the device. This ensures reliable user input for navigating the on-device settings menu without phantom presses.
3. **UART Interrupt:** Manages bi-directional serial communication between the microcontroller and the computer. This interrupt captures incoming threshold update commands from the GUI and transmits live telemetry (voltage, current, and fault status) back to the computer.

Computer Interface and Virtualization

To provide a seamless user experience, a dedicated GUI was developed. This interface allows users to view live power metrics, check fault statuses, and configure safe operating thresholds (Over-Voltage, Under-Voltage, and Over-Current).

For simulation and testing purposes, the communication link between the simulated ATmega32 and the Python GUI was bridged using com0com, a software utility that creates virtual COM port pairs. This allowed the simulated microcontroller hardware to communicate directly with the local Python script via a virtual serial connection.

Testing and Validation

The system underwent rigorous functional testing to verify its protective capabilities and software synchronization. A video demonstration was recorded to document the successful satisfaction of the following five core test cases:

1. Is it cut off the lines if the voltage goes above the over threshold volt?
2. Is it cut off the lines if the voltage goes under the under threshold volt?
3. Is it cut off the lines if the voltage goes above the over threshold current?
4. Is it blocks the two edits option at one time?
5. Is the device itself got updated if the computer version got updated?

Conclusion

The Microcontroller-Based Voltage Protector Device successfully met all design requirements. By leveraging hardware interrupts on the ATmega32, the system achieves real-time monitoring of voltage and current using a VT and the ACS712 sensor. The integration of a Python-based GUI via virtual COM ports provides an intuitive command interface, while the mutually exclusive lock mechanisms ensure stable configuration. The successful completion of the five core validation tests proves the system is highly capable of isolating and protecting electrical appliances from severe power grid fluctuations.